

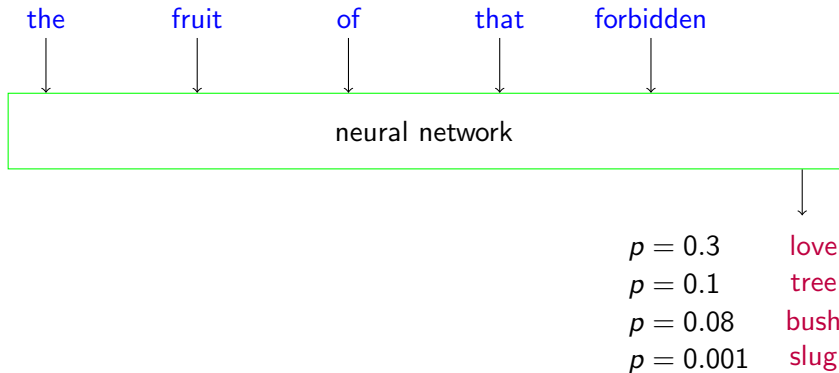
Large language models

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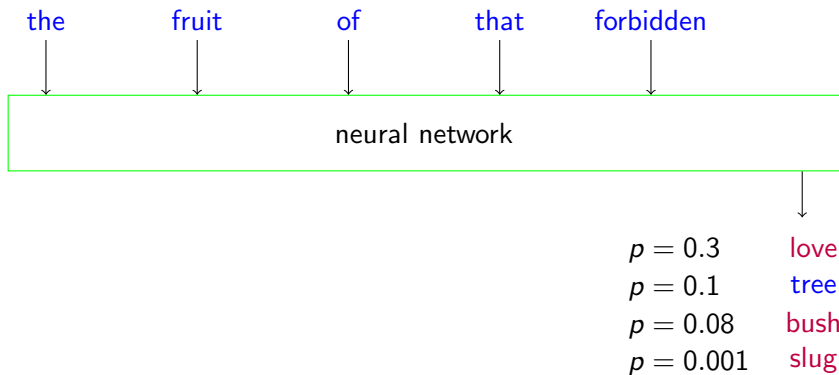
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Probabilities



Probabilities



Objective function

$$E = -\log p(\text{tree} | \text{and the fruit of that})$$

No contrastive terms

You might expect¹

$$E = -\log p(\text{tree}) - \log [1 - p(\text{love})] - \log [1 - p(\text{bush})] - \dots$$

but, unlike in word2vec, this isn't what's used. Remember though that first from a practical point-of-view the softmax provides a contrast:

$$p(w_i|\text{past}) = \frac{\exp z_i}{\sum_j \exp z_j}$$

and from a principled point-of-view the negative log-probability is an estimate of the cross-entropy, the cross-entropy in turn is the optimization-relevant component of the KL divergence.

¹conditional part of the probability left out for brevity

Next word prediction

The big advantage is that the training set doesn't need to be supervised, it sort of supervises itself, all that is needed is text, and lots of it.

Next word prediction

The claim is that a surprising number of tasks can be performed as next word prediction:

If I was asked 'is the film *Das Boot* about submarines' I would reply

and check if

$$p(\text{yes}) > p(\text{no})$$

Autoregressive generation

The main way we use next word prediction is in *autoregressive generation*, the idea is to add the predicted output to the input.

of mans first disobedience...

and = $\operatorname{argmax}_w(p[w|\text{of mans first disobedience}])$

of mans first disobedience and...

the = $\operatorname{argmax}_w(p[w|\text{of mans first disobedience and}])$

of mans first disobedience and the ...

fruit = $\operatorname{argmax}_w(p[w|\text{of mans first disobedience and the}])$

of mans first disobedience and the fruit...

...

Sampling

In fact, the *greedy* approach isn't usually used; just as the average face might look uncanny and unnatural, picking the most likely word makes text seem generic and odd. Instead the word is often sampled.

Sampling

random sampling: pick w with probability $p(w|\text{past } ws)$

top-k: throw away all but the k most likely words, normalize their probabilities and sample.

nucleus sampling: from the most probable downwards, keep enough words so that their probability adds to some value p_0 and then normalize and sample.

temperature sampling: change the softmax function to

$$\frac{\exp z_i / T}{\sum_j \exp z_j / T}$$

and sample.

Sampling

In some tasks you might then iterate forward a chunk of text multiple times and pick the best, this is, roughly speaking *beam sampling* though the way that this works is now much more complex, with parallel sampling and a trained judge or verifier.

Prompting

What we see:

"Why are there sausage dogs?"

What the LLM sees:

You are ChatGPT, an AI language model designed to assist users with a wide variety of tasks. Your role is to provide helpful, clear, and concise responses. Always maintain a neutral, polite, and professional tone; try to be nice.

You have been asked the question "Why are there sausage dogs?"

Your reply is . . .

The bit in blue is the *hidden prompt*; we may not know what the hidden prompt is, but it is *a lot* longer than that one! In fact, Anthropic does make the 'hidden' prompt public, it has lots of stuff, behavioural rules, definitions, instructions about tool use.

Prompting

The system prompt, the user's question, and whatever the machine reads from a web page or a document all arrive as the same thing: text. The model has no hard distinction between instructions and data, so text it retrieves can contain instructions it follows. This is *prompt injection*; it is the main security problem with agentic systems and nobody has a clean fix.

LLMs are vast

There are two ways LLMs are very big:

1. Extremely high-dimensional representation spaces.
2. Massive context windows.

Finetuning

The idea is that *pretraining* means training the machine on as much clean text as you can find, however, for a particular purpose you might want to train in a more directed way.

The obvious thing to do is just to train some more at the end using the specific corpus you are interested in. The problem is the number of parameters is huge!

Finetuning

A common strategy is *LoRA*, low rank adaptation, is to take the weight matrices, say $\mathbf{W}$ and add a low-rank matrix:

$$\mathbf{W} + \mathbf{AB}$$

where if $\mathbf{W}$ is an $n \times m$ matrix then $\mathbf{A}$ is $n \times r$ and $\mathbf{B}$ is $r \times m$ with r smaller than n and m . The extra training set is used to train $\mathbf{A}$ and $\mathbf{B}$.

Finetuning

Another strategy is an *adapter*; this means taking the output or inner layer of the machine and adding another set of layers trained to the specific task.

Reinforcement learning is sometimes used, this means humans rank outputs against criteria like 'politeness' and 'helpfulness' and this is used to create a reward model which is used for further finetuning.

These days reinforcement learning is more than a tool for fine tuning, it is baked into the whole process of training the network, for example, to encourage reasoning.

Comparison with humans

Transformers are trained on vast corpora, humans are not and yet humans, like machines, learn language. Are humans pretrained? If they are, in what sense are they pretrained?

Do humans learn the same thing?

Machines can now pass the Turing test; we have no idea what this means!

Reference: Jones & Bergen (2025, arXiv:2503.23674) ran a proper three-party test; GPT-4.5 with a persona prompt was judged human 73% of the time, more often than the actual humans.

What about humans?

Humans learn to speak using a much smaller corpus.

Language training in first-language learning is not preserved across cultures. At all.

Language learning

“whereas parents in white middle-class Anglo-American families look directly at a child learner and talk to it in the modified speech register often called parentese, in Kaluli families mothers carry the child in such a way that it is facing in the same direction as she is while answering on the child’s behalf in tripartite interactions using an unmodified version of Kaluli.”

Chomsky and all that

“The language faculty is a component of the human mind, physically represented in the brain and part of the biological endowment of the species.”

Reference: Chomsky, Noam. 1995. The Minimalist Program. Cambridge, MA: MIT Press.

The Universal Grammar

“If a Martian were to visit Earth, he would conclude that all human beings share the same language, with differences only at the level of detail.”

The Universal Grammar

Machines have inductive biases but no universal grammar, as far as we can tell.

Machines think!

translate “.....”

.....

The translated text from Morse code is: “there will be four burgers, aalong with cheezus crust. tea and biscuit, alal wake cheesus crust. tea and badass bikes... aall we’re tough one.... cowabunga! in the morning, roooz... floor zooo orr... faloor dorkwizard... sea air...”

Chain of thought prompting

Problem: John has 23 apples. Paul wants 20, but John only gives him half that. How many apples does John have left? **Solution:**

1. John starts with 23 apples.
2. Paul wants 20 apples, but John gives him **half** of that:

$$\frac{20}{2} = 10$$

3. So, John gives Paul 10 apples.
4. To find how many apples John has left, subtract the 10 apples he gave to Paul from the 23 apples he originally had:

$$23 - 10 = 13$$

Conclusion: John has **13 apples** left.

Chain of thought

Chain of thought reasoning started as a bolted-on addition to large language models which depended on prompts. Since o1/DeepSeek-R1 thinking is trained in: the model is RL-trained to produce a long hidden reasoning trace before answering, with the reward coming from verifiable outcomes, maths answers, unit tests, rather than human preference.

Fadhb na nÚll

Fadhb:

Tá fiche trí úll ag Aoife. Tá fiche ó Niamh, ach ní thug Aoife ach leath de sin di. Cé mhéad úll atá fágtha ag Aoife?

Réiteach:

1. Tá fiche trí úll ag Aoife.
2. Tá fiche ó Niamh, ach ní thug Aoife ach **leath** de sin di:

$$\frac{20}{2} = 10$$

3. Mar sin, thug Aoife 10 n-úll do Niamh.
4. Chun a fháil amach cé mhéad úll atá fágtha ag Aoife, bain 10 úll as an 23 úll a bhí aici ar dtús:

$$23 - 10 = 13$$

Conclúid: Tá **13 úll** fágtha ag Aoife.

Thinking in Python and English

Fine-Tuned Behavior: LLMs have access to tools like a Python execution environment, as in the code interpreter or Python sandbox feature. In these cases, the model is encouraged through hidden prompts and through training to utilize these tools for more accurate or dynamic responses. This behavior might be explicitly designed by developers to ensure the model uses available resources effectively when solving complex problems.

Arguing with itself

Multiple chains of thought are used to create different outcomes, these are compared, sometimes picking the majority, or most consistent answer or the machine 'argues', asking itself to compare the chains.

Distillation

It is almost impossible to think about how to combine two old-style neural networks; how to blend two weight matrices. However, now, LLMs can learn from each other: a network can be trained on the outputs of another, this is *distillation*. Most small open-weight models are distilled from large ones, and the large ones are partly trained on their own filtered outputs.

RAG!

Machines can improve their own responses by augmenting the prompt with facts, this is *RAG: Retrieval-Augmented Generation*; the LLM feeds information from a database into its own context window to improve responses.

Agentic AI!

Machines are now used as part of a complex environment with a tool use loop; the model calls tools, reads results, calls more tools: the system is the model plus the harness plus the tools plus memory plus other model instances, and the harness is now where a lot of the engineering happens. RAG is part of this, but it does more than feed a database into context; *context engineering* is the current jargon.

Machines think!

translate “Ships at a distance have every man’s wish on board”
into Irish

The translation of “Ships at a distance have every man’s wish on board” into Irish is:

“Tá mian gach fir ar bord long atá i bhfad i gcéin.”

Machines think!

how about Esperanto?

In Esperanto, the phrase “Ships at a distance have every man’s wish on board” can be translated as:

“Ŝipoj en la distanco havas la deziron de ĉiu viro surŝipe.”

Machines think!

How about Europeranto?

Eupanto is a playful, pseudo-language that blends elements from multiple European languages. Since Eupanto doesn't have strict grammar rules or a standardized vocabulary, translations can vary widely depending on the mix of languages used. Here's an attempt at translating "Ships at a distance have every man's wish on board" into Eupanto, using a blend of common European languages:

"Schiffe à la distanza haben jeder man's deseo a bordo."

Machines think!

The remarkable thing is that machines now seem to use language for thought, but it's English, or Irish, or French or, left to themselves, a mixture, nonetheless the internal use of language is part of how LLMs work. The relationship between language and machines is complex and fascinating!

Where is the intelligence!

There was a time that describing a neural network seemed simple, it was the matrix! Now the existence of the machine intelligence is more complex, it is **the matrix**, it is **the fine tuning layers**, it is **the prompt**, it is **a database**, it is a **set of tools**, it is **a harness**. It does not only have what it learned at training time, it has the messages it has created for itself in the conversation.

What is it like to Be a Transformer?

Well?